

Maxwell Kwesi Mac-Ocloo

East Lansing, MI, USA

915-273-1388 | macocloo@msu.edu

EDUCATION

Michigan State University, East Lansing, MI

Aug 2023 – May 2028 (Expected)

PH.D. BIOSTATISTICS

Advisor: Dr. David Todem

The University of Texas at El Paso, El Paso, TX

Aug 2021 – May 2023

M.SC. STATISTICS AND DATA SCIENCE

Advisor: Dr. Abhijit Mandal

University of Energy and Natural Resources, Sunyani, Ghana

Aug 2015 – Jul 2019

B.SC. STATISTICS (*Valedictorian*)

Advisor: Dr. Killian Asampana Asosega

RELEVANT COURSEWORK

Statistical Programming, Statistical Data Mining, Statistical Consulting (analysis and design), Generative Models, Spline Regression, Linear Regression Model, Mixed Model, Survival Analysis, Causal Inference, Mediation Analysis, Semi-parametric and Non-parametric Models.

SKILLS AND COMPETENCIES

Statistical Software & Programming: SAS, R Studio, Python Programming, SPSS, STATA, Microsoft Office

Soft skills: Effective verbal, written communication and interpersonal skills, time management skills, fast learner, team player, strong analytical skill.

EXPERIENCE

Biostatistics Intern

May 2026 – August 2026

Public Health Sciences, Henry Ford Health

Detroit, Michigan

- Collaborated with investigators and interdisciplinary research teams to formulate research questions, select appropriate statistical methodologies, interpret findings, and translate statistical results into scientifically meaningful conclusions.
- Contributed to scientific manuscripts and research reports by developing statistical methods sections, interpreting analytical results, synthesizing relevant literature, and preparing results for peer-reviewed dissemination.
- Designed and implemented statistical analyses for observational clinical and public health studies addressing treatment adherence, behavioral health, social support, substance use, and survival outcomes.

Data Analyst

June 2025 – April 2026

Department of Obstetrics, Gynecology and Reproductive Biology, MSU

East Lansing, Michigan

- Leveraged R and SAS for advanced statistical modeling—causal inference, hazard modeling, trend analysis and survival analysis—to examine prognostic determinants of breast cancer mortality across racial groups.
- Conducted rigorous statistical analyses to evaluate racial disparities in breast cancer survival and assess changes in mortality patterns across diagnosis periods.
- Developed clear reports, publication-quality visualizations, and statistical summaries to communicate findings to faculty and research collaborators.

Graduate Research Assistant

August 2023 – May 2025

Department of Epidemiology and Biostatistics, MSU

East Lansing, Michigan

- Applied causal inference and missing data methodologies to clinical and epidemiological datasets using R and SAS.
- Developed and implemented statistical models (e.g., sequential mean models, regression, survival analysis) to evaluate treatment effects in observational studies using SAS.
- Collaborated with interdisciplinary teams to translate statistical findings into actionable insights for health research.

Data Science Intern

June 2023 – August 2023

Lawrence Berkeley National Laboratory

Berkeley, California

- Managed and analyzed large-scale high-performance computing datasets to optimize system efficiency.
- Derived and visualized performance metrics using Python, demonstrating scalability in big data environments.
- Explored and derived performance metrics from large datasets and visualized performance metrics using Python.

Graduate Teaching Assistant

August 2021 – May 2023

The University of Texas at El Paso

El Paso, Texas

- Developed and delivered lesson plans tailored to diverse learning needs, reinforcing core mathematical concepts and problem-solving skills.
- Administered, graded, and tracked weekly assignments and quizzes for 40 students to monitor academic progress and provide timely feedback.
- Tutored 55 students in Proportion and Algebra through structured and innovative instructional approaches.

PUBLICATIONS

- Das, J.; Beyaztas, B.H.; **Mac-Ocloo, M.K.**; Majumdar, A.; Mandal, A. Testing Equality of Multiple Population Means under Contaminated Normal Model Using the Density Power Divergence. *Entropy* 2022, 24, 1189.
- Asosega, K. A., Opoku-Ameyaw, K., Otoo, D., **Mac-Ocloo, M. K.**, & Ayinzoya, R. (2020). Comparative Sensitivity Performance of the Discriminant Function and Logistic Regression under Different Training and Test Samples for Predicting Birth Outcomes. *Asian Journal of Probability and Statistics*, 6(3), 47-60.

PRESENTATIONS

- Exploring Real-time Data Transfers from Stanford Linear Accelerator Center (SLAC) to National Energy Research Scientific Computing Center (NERSC).
- Variance component testing in generalized linear models with random effect by Xihong Lin.
- Association between Social Support and Treatment Adherence in Head and Neck Cancer Patients.

ACADEMIC HONORS

- Valedictorian, University of Energy and Natural Resources 2019 Congregation
- Best student award in the School of Sciences, University of Energy and Natural Resources 2019 Congregation
- Best student award in the Department of Mathematics and Statistics, University of Energy and Natural Resources 2019 Congregation.

SERVICE & PROFESSIONAL AFFILIATIONS

Student Member
Vice President
Vice President
Vice President

ASA, SIAM
Biostatisticians and Epidemiologists at Michigan State, BEAMS (2026 – 2027)
African Graduate Student Association (AGSA), MSU (2026 – 2027)
Biostatisticians and Epidemiologists at Michigan State, BEAMS (2025 – 2026)